

Chapter 3

Methodology

3.1 Tools

Several external software tools evaluate and generate the updated DNP group parameters. In-house Python scripts collect and combine data, OpenMC generates the group parameters, and the MOOSE app Moltres evaluates and refines them at different fidelity levels [159, 136, 94]. Each tool is described in detail in its corresponding sub-section.

3.1.1 Python Scripts

The Python scripts used in the simplified DNP group generation rely on the SciPy, NumPy, Matplotlib, *pandas*, and Uncertainties packages [163, 60, 67, 156, 45]. These scripts automate the OpenMC sample irradiation simulation, collect data from the IAEA beta-delayed neutron emission database, combine OpenMC concentrations with half-life and emission probability data, and fit group parameters to the calculated delayed neutron count curve [40]. Section 3.2 provides further details on these scripts.

3.1.2 OpenMC

OpenMC is an open-source Monte Carlo neutron and photon transport code. Its Python Application Programming Interface (API) offers functionality that simplifies interfacing with in-house Python scripts. The sample irradiation process is straightforward and configurable, as users can adjust parameters through script inputs. OpenMC also offers useful feature in depletion, where a zero-power depletion step is rapidly calculated since only depletion is tracked. This feature allows resolving ex-core residence time depletion steps quickly. Additionally, OpenMC's transfer rate functionality during depletion enables simulation of chemical removal in an MSR.

For this work, the scripts use the Evaluated Nuclear Data File (ENDF)/B-VIII.0 based depletion chains provided by OpenMC and ENDF/B-VIII.0 cross section data. The thermal spectrum chain applies to neutron irradiation below 300 eV, while the fast spectrum chain applies to higher-energy neutron irradiation. Monoenergetic neutrons irradiate the fissile sample in the 0D flow model. This approach ensures the irradiation generates group parameters for a specific neutron energy rather than a particular spectrum.

3.1.3 Moltres

Overview

MOOSE is an open-source, parallel, finite element framework for multiphysics applications [56]. MOOSE includes many pre-built applications that couple physical phenomena. The HEAT CONDUCTION and NAVIER-STOKES modules are particularly useful in this work, as they can couple with Moltres for general thermal-hydraulic modeling [119, 93, 121, 117].

Moltres is an open-source MOOSE application that simulates MSRs by solving the multi-group neutron diffusion and thermal-hydraulics equations [94]. This work uses Moltres to simulate the MSRE with different DNP group parameters. Moltres also validates the updated DNP group parameters in this work.

The MSRE experimental data can be used for validation, and was previously modeled with Moltres [94]. The MSRE core consists of flow through channels, simplifying its modeling compared to open-pool designs due to the restricted flow path. Since the initial demonstration, Park et al. have noted general agreement in results, attributing differences to simplifications and assumptions in the geometry [117]. Additionally, Park has published a dissertation detailing 1-D, 2-D, and 3-D neutronic models of the MSRE [116].

Moltres can simulate transients using the MSR-specific DNP group parameters, and it can generate updated parameters with higher fidelity than the 0D method. Moltres couples three physical phenomena to model an MSR: neutron flux, DNPs, and thermal-hydraulics.

Flux Modeling

Neutron flux modeling determines the energy of neutrons interacting with the sample and the number of interactions. It determines the fission rate and the resulting fission product concentrations, including DNPs. The time-dependent multi-group neutron diffusion equation appears in Equation (2.27), recreated here:

$$\begin{aligned} \frac{1}{v_g} \frac{\partial \phi_g}{\partial t} = & \nabla \cdot D_g \nabla \phi_g + \sum_{g' \neq g}^G \Sigma_{s,g' \rightarrow g} \phi_{g'} \\ & + (1 - \beta) \chi_{p,g} \sum_{g'=1}^G (v \Sigma_f)_{g'} \phi_{g'} \\ & - \sum_{i=1}^I \chi_{d,g,i} \lambda_i C_i - \Sigma_{a,g} \phi_g - \sum_{g' \neq g}^G \Sigma_{s,g \rightarrow g'} \phi_g \end{aligned}$$

Modeling the neutron flux requires accounting for the DNPs. DNPs contribute to the neutron flux and possess their own energy spectra, necessitating their inclusion as a separate term.

Precursor Modeling

The time dependent DNP concentration is given in Equation (2.28) and recreated here:

$$\frac{\partial C_i}{\partial t} = \beta_i \sum_{g=1}^G (v \Sigma_f)_g \phi_g - \lambda_i C_i - \nabla \cdot (C_i \vec{u}) + \nabla \cdot \frac{v_T}{Sc_T} \nabla C_i$$

An issue arises with this equation because Moltres generates MSR-specific DNP group parameters, yet the equation requires predefined DNP group parameters. To generate the DNP group parameters, this equation is replaced by (2.51), recreated here:

$$\frac{dN_j}{dt} = S(\vec{r}, t) - \lambda_{net,j} N_j(\vec{r}, t) + \nabla \cdot (D_{mix} \nabla N_j(\vec{r}, t) - \vec{u}(\vec{r}, t) N_j(\vec{r}, t))$$

Specifically, the solution to this problem is to use spatially resolved depletion over the irradiation simulation; instead of tracking concentrations of DNP groups, the simulation tracks concentrations of the individual DNPs.

Thermal-Hydraulic Modeling

Moltres offers time-dependent, incompressible, buoyancy-driven flow using Equations (3.1) and (3.2) recreated from [116]:

$$\rho \frac{\partial \vec{u}}{\partial t} = -\rho(\vec{u} \cdot \nabla) \vec{u} - \nabla p + \mu \nabla^2 \vec{u} + \rho \alpha \vec{g} (T - T_{ref}), \quad (3.1)$$

$$\nabla \cdot \vec{u} = 0, \quad (3.2)$$

where,

ρ = fluid density

p = pressure

μ = dynamic viscosity

α = coefficient of thermal expansion

$\vec{g}$ = gravitational force vector

T = fluid temperature

T_{ref} = reference temperature at which the nominal density is provided.

The temperature distribution is given by Equation (3.3) recreated from [116], which shows the temperature advection-diffusion as:

$$\rho c_p \frac{\partial T}{\partial t} = -\rho c_p \vec{u} \cdot \nabla T + \nabla \cdot (k \nabla T) + Q_f - Q_s, \quad (3.3)$$

and by Equation (3.4) recreated from [116], which gives the fission heat source term as:

$$Q_f = \sum_{g=1}^G \epsilon_g \Sigma_g^f \phi_g, \quad (3.4)$$

where,

c_p = specific heat capacity of molten salt

k = effective thermal conductivity of molten salt

Q_f = fission heat source

ϵ_g = average fission energy released by neutrons in group g ,

Q_s = heat sink/removal.

With the combination of the flux modeling, precursor modeling, and thermal-hydraulic modeling, Moltres can almost simulate MSRs. However, there is one additional MSR phenomena to complete the model. This phenomena is the flow of the fuel salt between the in-core to ex-core regions.

Ex-core and In-core Coupling

Moltres treats the ex-core region as a tightly-coupled 1D pipe geometry with coupling using Picard iterations and the inlet/outlet boundary conditions. The specific equations to treat the variables of interest in this geometry can be derived from their full forms. For example, the flux and eddy viscosity are zero in the ex-core region, leading to Equation (3.5), which describes the precursor concentration in the ex-core region as:

$$\frac{\partial C_i}{\partial t} = -\lambda_i C_i - u_x \frac{\partial C_i}{\partial x}. \quad (3.5)$$

3.2 DNP Group Generation

Generating DNP group parameter data in a reactor-agnostic manner requires a reactor-agnostic generation process. Fully detailing the flow regime with a large number of parameters is computationally infeasible. Instead, the flow is simplified to two parameters: an in-core average residence time, τ_{in} , and an ex-core residence time, τ_{ex} .

This approach captures fuel salt advection around the primary loop and approximates turbulent and diffusive mixing. The terms derive from the averaged behavior of a fuel salt control volume, treating thermal-hydraulic effects, such as vortices, as changes in residence times. This method allows for approximate fitting of any circulating fuel reactor, requiring only the generation of relevant parameters for determining the associated DNP group parameters.

The other required parameters describe chemical removal, which varies by fission product and reprocessing rate. Refueling is not considered in this work, as its effects are negligible over the short time frames analyzed. An irradiation time of approximately seven minutes suffices to reach equilibrium DNPs concentrations, as the saturation time frame is on the order of multiple half-lives of the longest-lived DNP. Elements with cycle times on the order of months to years exhibit negligible changes, as discussed in Section 4.1.6. Elements with changes on the order of seconds to days contribute sufficiently and must be included for accurate data generation.

The MSBR reprocessing scheme serves as the basis for cycle times in this work, as shown in Table 3.1. Rare earths, semi-noble metals, volatile fluorides, discard, and higher nuclides have negligible effects over the time scales studied. Section 4.1.6 examines these longer cycle times to confirm their negligible impact. Section 4.3 scales the cycle times to analyze how reprocessing rates influence the group parameters.

Table 3.1: MSBR Online Reprocessing Cycle Times from [133].

Reprocessing Group	Element(s)	Cycle Time (Full Power)
Rare Earths	Y, La, Ce, Pr, Nd, Pm, Sm, Gd	50 days
Rare Earths	Eu	500 days
Noble Metals	Se, Nb, Mo, Tc, Ru, Rh, Pd, Ag, Sb, Te	20 seconds
Seminoble Metals	Zr, Cd, In, Sn	200 days
Gases	Kr, Xe	20 seconds
Volatile Fluorides	Br, I	60 days
Discard	Rb, Sr, Cs, Ba	3435 days
Salt Discard	Th, Li, Be, F	3435 days
Protactinium	Pa	3 days
Higher Nuclides	Np, Pu	16 years

The parameters used include incident neutron energy, target fissile nuclide, in-core residence time, ex-core residence time, reprocessing scheme, and reprocessing scaling vector. Each unique parameter adds a dimension to the data, exponentially increasing the computational cost of generating group parameters for every combination. This work uses the MSBR reprocessing scheme with a single scaling value applied to each reprocessing group instead of a unique scaling value for each. The incident neutron energies that irradiate the sample are 0.0253 *eV* and 1 *MeV*. Section 4.3 determines the minimum number of residence times and their values required to capture general trends.

3.2.1 0D Flow Model

A simplified model is necessary to generate sufficient data for each combination of input parameters. This model estimates the magnitude of variation and identifies general trends. It treats the flow in a zero-dimensional manner using in-core and ex-core residence times. The fission history function is a square wave, with a constant fission rate during the in-core residence time and no fission during the ex-core residence time. This approximation simplifies the analytical form of the fission history for creating the non-linear least-squares problem, as shown in Appendix A.

Figure 3.1 shows how OpenMC uses this fission history to simulate irradiation on a control volume of fuel salt circulating in the primary loop. The model passes eight parameters into OpenMC [136]: in-core residence time, ex-core residence time, neutron irradiation energy, target fissile nuclide, target density, total irradiation time, temperature, and reprocessing terms. Three parameters, shown in green, remain constant: temperature, target density, and total irradiation time. The remaining inputs, shown in yellow, vary to generate new group parameters. Section 3.3 provides further details on this process.

After OpenMC completes the fixed-source simulation, the final nuclide concentrations in the irradiated sample are saved for analysis in custom scripts. These scripts combine the concentrations with data from the IAEA beta-delayed neutron emission database to generate a delayed neutron count rate [40]. A non-linear least squares fitting function from SciPy then calculates the group half-lives and yields using the Trust Region Reflective algorithm

[164, 22]. These group parameters correspond to a specific nuclide, neutron energy, in-core residence time, ex-core residence time, reprocessing scheme, and reprocessing scalar. Section 3.4 provides further details on this block. Some additional factors to consider include self-shielding due to sample density (while avoiding self-multiplication), temperature-induced cross-section variations, and under- or over-irradiation, all of which can affect the results [76].

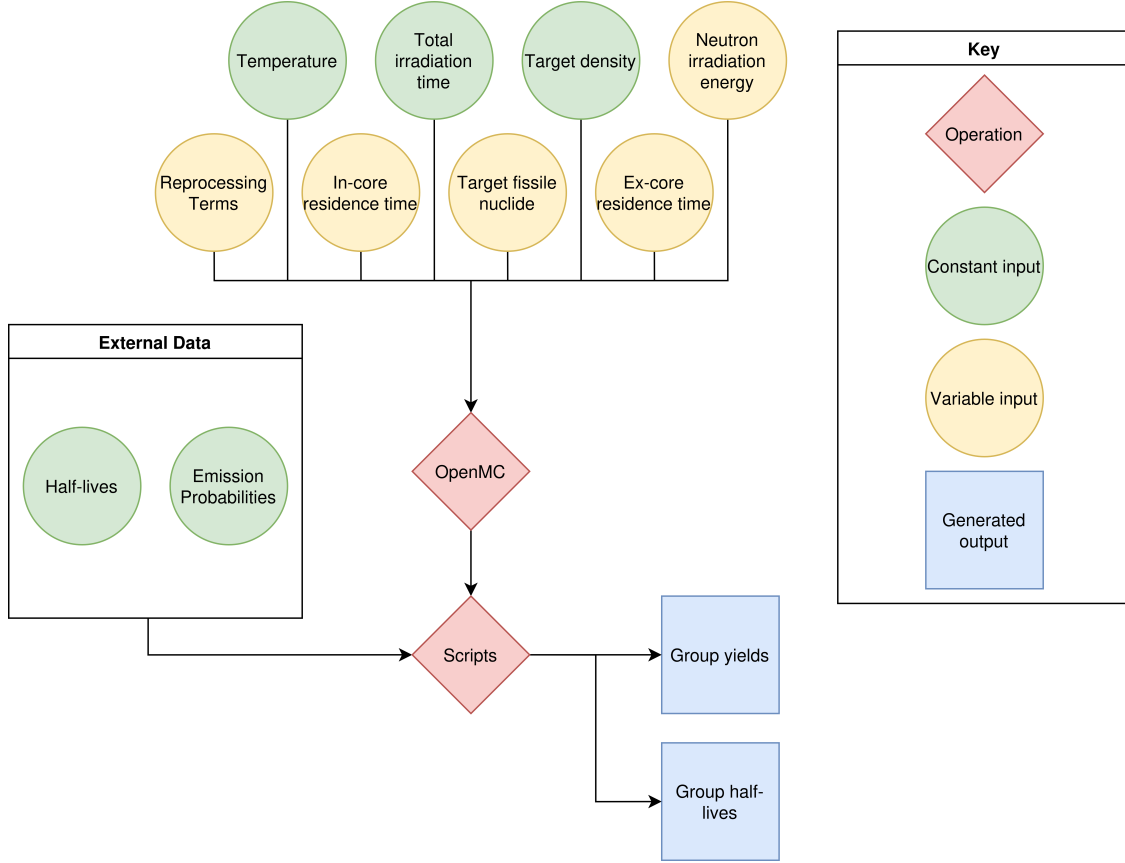


Figure 3.1: Flow diagram of 0D flow model used to generate flow-inclusive precursor group fit parameters.

The OpenMC model uses a three-gram spherical sample with vacuum boundary conditions. It treats the problem as a fixed-source simulation, with a uniform neutron source distributed evenly throughout the sphere. This approach is not physical, as a real neutron source would have a spectrum and originate externally. However, the exact neutron spectrum for irradiating samples to generate DNP group parameters is not provided in the literature. The exact neutron energy spectrum is not needed, as only significant differences in the neutron energy spectrum affect the DNP group parameters. The source distribution also has negligible impact, as the sample is small and exhibits minimal self-multiplication. Thus, this approximation does not limit the effectiveness of the 0D flow model.

Another consideration for the OpenMC model is maintaining a consistent source particle rate. While not strictly necessary for group parameter generation since the values are per-fission, it simplifies the data analysis, as the tallies depend on the source particles. Equation (3.6) provides the source rate for the irradiation simulations:

$$\dot{S} = S_{net} \frac{\tau_{in} + \tau_{ex}}{\tau_{in} T}, \quad (3.6)$$

where,

$$\begin{aligned}\dot{S} &= \text{the source particle rate } [\frac{n}{s}] \\ S_{net} &= \text{the total number of source particles } [n] \\ \tau_{in} &= \text{the in-core residence time } [s] \\ \tau_{ex} &= \text{the ex-core residence time } [s] \\ T &= \text{the net irradiation time } [s].\end{aligned}$$

3.2.2 Spatially Resolved Model

The 0D flow model rapidly generates DNP group parameters, but it simplifies the treatment of flow. Instead of modeling the actual fuel salt flow, it estimates fuel circulation using residence times for each region. Figure 3.2 shows the flow diagram for a spatially resolved model using Moltres instead of OpenMC.

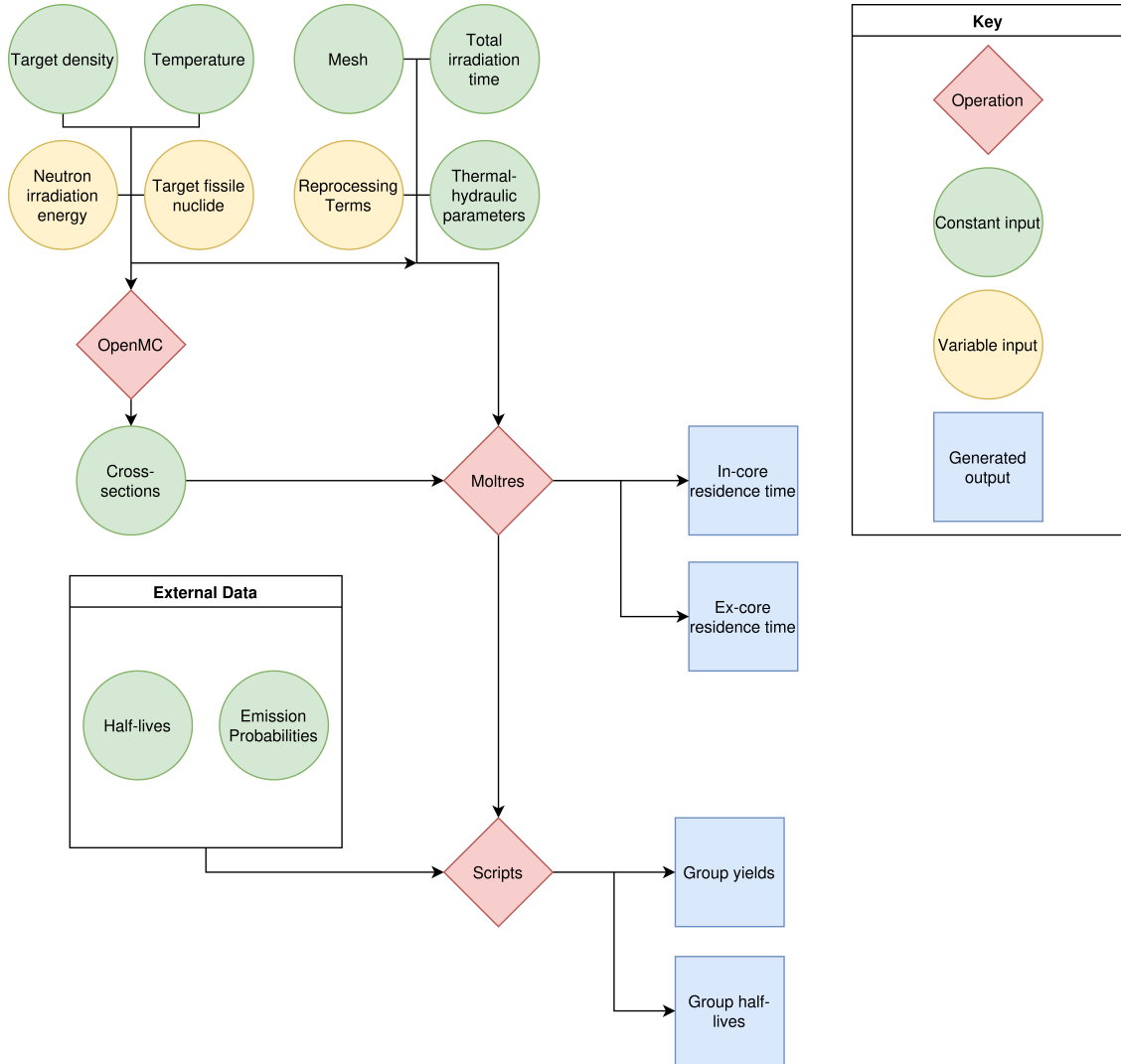


Figure 3.2: Flow diagram of spatially resolved flow model used to generate flow-inclusive precursor group fit parameters.

The spatially resolved flow model introduces two new parameters absent in the 0D flow model. The first is the mesh, which defines the problem geometry. MOOSE provides built-in meshing tools, though complex geometries may require external tools such as Gmsh [55]. The second is the thermal-hydraulics parameters, which capture physical phenomena enabled by spatial resolution, as discussed in Section 3.1.3.

In the spatially resolved model, the in-core and ex-core residence times are outputs rather than inputs. These times derive from the movement of the irradiated sample over time in the fuel salt. The in-core and ex-core residence times vary as the irradiated sample diffuses and mixes further into the fuel salt. The net residence time, τ_{net} , remains constant unless the model changes, such as an increased temperature difference or induced flow rate. Equation (3.7) defines the in-core residence time as:

$$\tau_{in}(t) = \frac{\int_{V_{in}} \int_0^t N(\vec{x}, t) dt d\vec{x}}{\int_V \int_0^t N(\vec{x}, t) dt d\vec{x}} \tau_{net}, \quad (3.7)$$

where

V_{in} = the in-core volume [cm^3]

V = the total primary volume [cm^3]

$N(\vec{x}, t)$ = the time and space resolved sample concentration [$\frac{atoms}{cm^3}$]

τ_{net} = the net residence, or circulation, time of the fuel salt [s].

The ex-core residence time can be calculated in the same way, or alternatively as shown in Equation (3.8) as:

$$\tau_{ex} = \tau_{net} - \tau_{in}. \quad (3.8)$$

3.3 Delayed Neutron Count Rate Generation

Although the 0D flow model overview has been discussed, execution details were not. The following sub-sections provide specific information on the methodology behind the OpenMC block. This includes a detailed discussion on the concentrations generated by OpenMC and the data used with those concentrations to produce a delayed neutron count rate.

3.3.1 DNP Concentrations

The OpenMC model uses a fixed-source problem on a three-gram spherical sample of a fissile target. A uniformly distributed source is placed inside the target with a vacuum boundary condition at the edge of the sphere. The sample must have a sufficiently large volume to ensure a high number of fissions, but excessive volume increases self-shielding and self-multiplication. Adjusting the density mitigates this issue: higher density increases self-shielding and self-multiplication, while lower density reduces fission probability and worsens statistical accuracy.

The fixed-source in this model consists entirely of neutrons at a specific energy, rather than a spectrum. This energy dictates fission cross sections, transmutation cross sections, and fission yields. Traditionally, tabulated group parameter data is formatted for a single energy or energy range, such as fast or thermal. However, macroscopic experiments, discussed in Section 2.1.1, use reactors with varying neutron energy spectra, not mono-energetic neutrons. This could be addressed by modeling a reactor in OpenMC and using a k-eigenvalue solve to generate a neutron energy spectrum. However, this work assumes the spectral effects are sufficiently small to exclude. Photons

can also generate delayed neutrons through photofission, though this is not investigated here [78].

The neutron source distribution does not replicate the experimental process, where neutrons irradiate the sample externally rather than internally. However, an evenly distributed source eliminates self-shielding concerns. The exact spatial locations of fission events within the sample should negligibly affect the nuclide concentrations, which are the outputs of the OpenMC simulation. With a density of $10 \frac{g}{cc}$, the spherical sample has a radius of approximately 0.4 cm . The spatial resolution of concentrations within the sample is insignificant. Additionally, the concentrations extracted from the OpenMC simulation are spatially collapsed in the post-processing, so the spatial effect of a uniformly distributed source serves only to reduce any self-shielding effects which could reduce the statistics and cause more transmutation than expected.

After running the OpenMC simulation, the concentrations are collected from the depletion results for the final irradiation time step. These final concentrations are passed to other scripts that convert the concentrations into a delayed neutron count rate.

3.3.2 Concentrations to Count Rate

After receiving the concentrations from OpenMC, the scripts determine the concentration of each nuclide as a function of time during the measurement period. The concentrations are calculated based on half-life, as shown in Equation (3.9), where the concentration as a function of time is:

$$N(t) = N_0 e^{-\lambda t}. \quad (3.9)$$

This simple equation does not account for decay chains between nuclides. An alternative approach uses repeated zero-power depletion simulations in OpenMC to generate concentrations as a function of time, incorporating decay chains. These concentrations are available at coarse time steps and linearly interpolated using SciPy, allowing the scripts to access more accurate concentrations at any time after irradiation [162]. This approach only models delayed neutrons generated during decay, as it is beyond the scope of this work to model other particles [28]. Section 4.1.7 compares these two approaches.

The concentrations are combined with Equation (2.2), recreated here:

$$\dot{n}_d(t) = \epsilon \sum_{k=1}^K P_{n,k} \lambda_k N_k(t).$$

The efficiency term, ϵ , is set to 1 in this work, as the simulated method includes every produced delayed neutron in the count rate. The contributions of each DNP over time are summed to generate the delayed neutron count rate.

3.4 DNP Group Reconstruction

Once the delayed neutron count rate is generated, it is converted into DNP group parameters. Fitting the group parameters requires a non-linear least squares method, as both the group half-lives and yields must be determined. The equation describing the problem appears in Equation (2.3), recreated here:

$$\dot{n}_d(t) \approx \epsilon F_s \sum_{k=1}^K \bar{v}_{d,k} \lambda_k e^{-\lambda_k t}.$$

An important consideration for this process is the selection of time values for generating the group fits. Increasing the number of time nodes raises computational costs, but too few nodes result in a poorer fit [114]. Section 4.1.5

shows the sensitivity study for the time nodes, for both the node spacing and the maximum time to measure delayed neutrons.

Another consideration for the non-linear least squares solve is the covariance, which is larger for the pulse irradiation fit than for the saturation irradiation fit. This results from multiplying the group yield by the group decay constant. Attempts to replace this multiplication with an independent term produce poor fits, as the multiplication is not independent and is necessary for accurate modeling.

The non-linear least squares process also involves linearization and normalization of the data. The normalization is handled by taking the delayed neutron count rate, dividing by the fission term ¹, and taking the natural logarithm. The normalized count rate is then passed into the non-linear least squares fitting. The linearization is handled by taking the natural logarithm of the delayed neutron count rate. This treatment reduces both the covariance and the χ^2 of the fit. Bounds from 0 to 1000 are set for each term to speed up the process, while convergence tolerances are set slightly above machine tolerance at 2.23×10^{-16} to ensure a well-refined solution. The convergence criteria include changes in the independent variables and the norm of the gradient, with termination occurring when either criterion meets the set tolerance.

Sample self-multiplication, mentioned in Section 2.1.1, is another consideration. However, since the fission counts are taken directly from OpenMC tallies, self-multiplication is included and does not introduce error.

The final consideration in DNP group reconstruction is uncertainty. This work tracks uncertainties using the uncertainties package [45]. These uncertainties include uncertainties in cross sections, fission yields, half-lives, and emission probabilities. These uncertainties are used in the χ^2 minimization, as shown in Equation (2.6).

Once the group fits and their uncertainties are generated, the final step to obtain the finalized group parameters is to combine the saturation and pulse irradiation results. Saturation irradiation provides better statistics for longer-lived groups, while pulse irradiation is better for shorter-lived groups. Brady combines the results by using four long-period groups from saturation irradiation and four short-period groups from pulse irradiation. The groups are renormalized at the second group, where the two longest-lived group half-lives and yield ratios use the saturation results [21]. The remaining groups use the pulse irradiation results.

This work uses a similar but slightly simplified approach. The group parameters are first calculated for both pulse and saturation irradiation. Then, the two longest-lived groups use the saturation irradiation results, while the four shortest-lived groups use the pulse irradiation results.

3.5 Impact of DNP Groups on Transients

The impact of the DNP groups on transients is evaluated using Moltres with fluid-flow-velocity field coupled multi-group diffusion kinetics, as discussed in Section 2.1.6. This approach provides high-fidelity results by coupling thermal-hydraulics and neutronics, a necessity for MSR modeling. Modeling every DNP in space and time would be prohibitively expensive, so they are instead modeled using DNP group parameters. Moltres allows direct comparison between static DNP group parameters, which do not account for MSR phenomena, and parameters that do.

The results of interest from this analysis include the reactivity and power responses to various initiating events. The initiating events and transients vary, as different reactor responses are studied. The events of interest depend on the perspective from which the reactor response is analyzed.

Specifically, there are four different perspectives of interest in this work: validation, safety, security, and safeguards. Each perspective involves a unique combination of motivation, proposed initiating event, and investigated response. These are discussed in detail in their respective sections.

¹The fission term is either the net number of fissions or the fission rate, depending on the irradiation type.

3.5.1 Validation

The motivation for validation is to confirm that the proposed methodology and results align with experimental data. This ensures the demonstrated updated DNP group parameters are applicable to actual MSRs. Discrepancies between experimental results and the model highlight potential weaknesses in the approach. If no discrepancies exist, confidence in the results increases for problems similar to those validated.

The validation of the DNP group parameters follows a multi-stage approach. The first stage compares the irradiated sample's delayed neutron count rate to the Keepin et al. experiment from 1957 [76]. If the count rates do not match, the issue lies in the data or the aggregation of individual DNPs into the count rate. If the count rate is accurately predicted, the formulation is considered valid. The second stage validates the generated DNP group parameters using MSRE pump start-up and coast-down experimental data to assess their accuracy in modeling reactivity changes from DNPs [126]. This experiment has been studied using Moltres and Quasimolto, allowing comparison with their results [129, 116]. However, previous models excluded the inlet and outlet plena, leading to discrepancies with experimental data. A model including the plena, combined with updated DNP group parameters, should yield results closer to the experimental data.

3.5.2 Safety

The motivation for studying safety is to protect the public and meet regulatory requirements. Safety is discussed in detail in Section 2.3.1, though the general takeaway is that many safety-related transients exist. The DNP group parameters are critical in these transients, as they influence the reactor period, power, and temperature.

The transients investigated are ULOHS, ULOF, UTOP, and FSOC. For each transient, the reactor period, power response, maximum temperature, minimum temperature, and reactivity response are recorded. The reactor period is critical, as it reflects the reactor's controllability. The power response shows how changes in the DNP group parameters affect other responses, such as temperature. It is important to make sure differences in the DNP parameters don't lead to temperature changes or oscillations which result in unsafe reactor conditions.

3.5.3 Security

Security is discussed in detail in Section 2.3.2, but the general takeaway is that bad actors may attempt to sabotage the reactor. This ties directly into the safety design study, as sabotage aims to create unsafe reactor conditions. Ensuring the MSR operates safely after sabotage is critical. If static DNP group parameters predict safe operation while updated parameters predict unsafe operation, correcting this discrepancy in the reactor design is essential.

Since the safety study covers most transients relevant to sabotage, additional considerations can be incorporated into these transients. As sabotage may initiate multiple safety-related transients simultaneously, it is reasonable to investigate such scenarios. With four unique transients, this results in six double-transient combinations, three triple-transient combinations, and one quadruple-transient combination, totaling ten security-related sabotage-initiated transients.

3.5.4 Safeguards

Safeguards are discussed in Section 2.3.3, but the main takeaway is the importance of preventing SNM diversion in MSRs. Detection methods, such as neutron detectors and surveillance systems, play a key role in safeguarding the reactor [175]. The updated DNP group parameters in this work affect the effective delayed neutron fraction and reactivity response. Using static DNP group parameters to detect SNM diversion predicts less accurate reactivity

responses than the updated parameters. If the difference is significant, relying on static parameters may reduce the detection probability of a diversion scenario.

To investigate this, diversion scenarios are studied with a single significant quantity of material removed over the course of one year, simulated using OpenMC. The DNP group parameters are then generated for the fuel composition at that time, weighted by the volumetric fission rate fraction of each fissile nuclide, F , similar to Equation (2.10). For example, the total delayed neutron yield is given as:

$$\bar{\nu}_d = \left(\sum_F \int \Sigma_f^{(F)} \phi \, dE d\vec{r} \right)^{-1} \sum_F \left(\bar{\nu}_d^{(F)} \int \Sigma_f^{(F)} \phi \, dE d\vec{r} \right), \quad (3.10)$$

where,

$$\bar{\nu}_d^{(F)} = \text{the total delayed neutron yield for nuclide } F \text{ [-]} \quad (3.11)$$

$$\Sigma_f^{(F)} = \text{the macroscopic fission cross section for nuclide } F \text{ [cm}^{-1}\text{]}. \quad (3.12)$$

These weighted group parameters are generated using the static and the updated values. Then, they can be used in simple step reactivity insertion transients, where the following four sets of data are compared: static group parameters without diversion, static group parameters with diversion, updated group parameters without diversion, and updated group parameters with diversion. If the difference between the static group parameters with and without diversion is smaller or similar to the difference between the static and updated group parameters, then this means detection of that diversion scenario requires models to account for the MSR physical phenomena when generating group parameters.